

Nithan B S

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Summary

Environmental Science graduate transitioning into Climate Risk and Environmental Data Analytics, combining domain knowledge in environmental science with hands-on experience in Python, machine learning, geospatial analysis and sustainability-focused analytics. UGC-NET qualified in Environmental Sciences, with prior research exposure in multi-hazard early-warning systems and independent projects spanning climate-risk modelling concepts and ESG/BRSR analytics.

Technical Skills

- **Environmental & Climate:** Climate Risk, Physical & Transition Risk, Climate Data Analysis, CMIP6, RCP/SSP Scenarios, Environmental Data Processing.
- **GIS & Geospatial:** GIS, GeoPandas, Geospatial Analysis, Spatial Visualisation, Hazard Mapping.
- **Climate Data:** Python, xarray, rioxarray, NetCDF, Pandas, NumPy
- **Data & Analytics:** SQL, Scikit-learn, Power BI.
- **ML/NLP:** PyTorch, Transformers, NLP, BERT

Research Experience

Research Intern — Regional Integrated Multi-Hazard Early Warning System (RIMES)

Chennai | May – July, 2019

- Conducted a **district-level drought-risk assessment for Dharmapuri, Tamil Nadu**, integrating Landsat-derived **NDVI/NDWI, SPI, rainfall anomalies and Moisture Adequacy Index (MAI)** to assess meteorological and agricultural drought conditions.
- Applied **GIS-based spatial and temporal analysis** to map drought patterns across blocks and identify water-limiting areas, including Pennagaram and Nallampalli.
- Examined relationships between **rainfall anomalies and vegetation response**, and documented methodological limitations including the need for longer precipitation baselines for statistically robust SPI analysis.

Selected Projects

ESG / BRSR Intelligence Platform - Retrieval-Augmented Generation (RAG) Pipeline

Python | NLP | ESG Analytics | Streamlit

- Developed an analytical platform for extracting and evaluating ESG/sustainability disclosures from Indian corporate BRSR reports.
- Applied NLP and structured-data techniques to transform unstructured sustainability disclosures into comparable ESG indicators.
- Explored automated identification of disclosure gaps and sustainability metrics to support ESG assessment and reporting analysis.

ENSO-IOD Teleconnections & Bushfire Risk

Climate Research | GIS | Geospatial Analysis

- Reviewed research spanning the development of the ENSO-IOD teleconnection and its potential relationship with climate extremes.
- Examined evidence concerning positive IOD events and Australian bushfire conditions.
- Used geospatial visualisation to examine the spatial anomalies occurring between different climate phenomena and reported extreme disaster patterns.

Career development & Research | 2020 - Present

- Qualified UGC-NET in Environmental Sciences (2021) and cleared PhD entrance examinations in Environmental Sciences.
- Participated in competitive selection processes for scientific/research positions, including reaching the NIOT Scientist-B interview stage.
- Continued independent development in Python, data science, ESG analytics, geospatial analysis and climate-risk research alongside examination preparation.

Education

Integrated M.Sc., Environmental Sciences (2015-2020)

Central University of Rajasthan

Certificates

- UGC – NET Environmental Science
- Master Data Science: GUVI - HCL
- Generative AI Workshop: GUVI - HCL